

Error Prone PCR (EP-PCR)

Introduces random mutations to genes (or gene segment) of interest, creating a library of mutated DNA. Screening of this mutated DNA library can help determine whether random mutations at certain areas changed function (introduction of new function, deletion of function, or alteration of function) of the gene. This can also be used as an vitro technique to mimic natural mutation but with more control, and at a faster pace.

Notes before starting:

- Time required to determine the amplification efficiency of EP-PCR is 3 hours
- Optimizing conditions to improve EP-PCR efficiency requires 1 day (optional)
- Performance of the EP-PCR protocol can take 2hours to 1 day depending on cycles performed
 - If periodic gel purification is needed, then 1 extra day is needed
- Cloning a sample of EP-PCR product may be carried out in 1 day
- Sequencing of DNA to determine the level of mutagenesis may require 1-5 days
- Analysis of sequence and results and calculating the degree of mutagenesis takes 1 day

Reagents

✓	Name	Quantity	Location/Notes
☐	Tris Cl (pH = 8.3)	100mM	
☐	KCl	2M	
☐	MgCl ₂	200mM	
☐	dCTP (pH = 7)	25mM	
☐	dTTP (pH = 7)	25mM	
☐	dATP (pH = 7)	5mM	
☐	dGTP (pH = 7)	5mM	
☐	5' PCR primers	100μM	
☐	3' PCR primers	100μM	
☐	DNA template (400 bp in length)	200 pg/μl	This may be variable depending on how long

			our DNA is* - Potentially changes the quantity of everything else our DNA is not 400 bp
☐	MnCL ₂	25 mM	
☐	Taq DNA Polymerase	5 U/μl	

Equipment

✓	Name	Quantity	Location/Notes
☐	100 μl PCR tubes		
☐	Thermal Cycler		
☐	TOPO T/A cloning kit		Double check what this is, if we can replicate without buying kit
☐	QIAprep kit (Qiagen)		Same as above
☐	Agarose gel (refer to electrophoresis protocol)		

Instructions

1. Make the following **PCR reaction mixture** in a 100- μ l PCR tube **on ice**
 - Note: reagent names have been shortened to just the chemical name for simplification, refer to reagents table for exact molar concentration. Unless specified, molar concentration is the exact same as seen in the table.
 - a. 51 μ l water
 - b. 10 μ l Tris
 - c. 2.5 μ l KCl
 - d. 3.5 μ l MgCl₂
 - e. 4 μ l dCTP
 - f. 4 μ l dTTP
 - g. 4 μ l dATP
 - h. 4 μ l dGTP
 - i. 2 μ l 5' Primer
 - j. 2 μ l 3' Primer
 - k. 10 μ l template DNA
 - l. 2 μ l MnCl₂
 - i. **IMPORTANT:** MnCl₂ should be added *immediately* before thermal cycling initiation.
 - m. 1 μ l DNA polymerase

Total volume: 100 μ l
2. **Add Taq polymerase** when thermal cycling reaction has reached the **first annealing** step
 - a. Note: First annealing is when the DNA strand *anneals/primers bind* for the first time
3. Place tube in thermocycler and perform ~12 PCR cycles (or until 1000-fold product input template/10 doublings achieved)
 - a. Note: Starting conditions vary. Annealing temperatures should be kept >50°C if possible to avoid mispriming.
 - b. Reasonable conditions to start at:
 - i. **Denaturation:** 94°C for 1 minute
 - ii. **Annealing:** 60°C for 1 minute
 - iii. **Extension:** 72°C for 3 minutes
4. Run ethidium bromide-containing agarose gel to confirm the amount and correct molecular weight of the product
5. Clone and sequence a sample of resulting PCR DNA to determine the frequency of mutations in the product using TOPO T/A cloning kit and QIAprep kit
 - a. *Note: Double check if these kits are necessary, and if there have been improvements to the protocol since article publication date (2001)

Alternate Protocol: Mutagenizing a Library of Sequences

Note: If information on this topic is needed, follow the article referenced at bottom of the document. If this protocol is needed, further research and protocol writing can be done.

Notes

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References

<https://currentprotocols.onlinelibrary.wiley.com/doi/10.1002/0471142727.mb0803s51>